Module 10: Epigenetics — Study Questions

1. What does "gene expression" mean? Why don't all cells express all genes at all times?
2. Every cell in your body has the same DNA. Explain how a muscle cell and a nerve cell can look and function so differently.
3. What is a transcription factor? How do activators and repressors control gene expression?
4. What is a promoter? What is an enhancer?
5. What is an operon? Describe how the lac operon works when lactose is absent vs. present.
6. Compare inducible operons (lac) vs. repressible operons (trp). How do they differ in their default state?
7. List the five levels at which gene expression can be regulated (from chromatin to post-translational).
8. What is alternative splicing? How can one gene produce multiple different proteins?
9. Define epigenetics. How does it differ from a genetic mutation?
10. What is DNA methylation? Does it typically activate or silence gene expression?
11. What are histones? How does histone acetylation affect gene expression?
12. Explain the difference between euchromatin and heterochromatin. Which allows gene expression?
13. What is X-inactivation? Why does it occur? Explain how calico cats demonstrate it.
14. Define point mutation vs. frameshift mutation. Why are frameshift mutations typically more damaging?

15. How can mutations in tumor suppressor genes or proto-oncogenes lead to uncontrolled cell division?

16. Give two examples of how environmental factors can influence gene expression through epigenetic mechanisms.

17. Identical twins share the same DNA. Why might they develop different traits or disease risks as they age?

18. A researcher discovers that a tumor suppressor gene in cancer cells has heavy methylation in its promoter region but no mutations in its DNA sequence. Explain how this could contribute to cancer.

19. Compare the lac operon model (prokaryotic) with enhancer/transcription factor regulation (eukaryotic). What fundamental principle do they share?

20. A student says: "Epigenetics proves that Lamarck was right — organisms can pass on acquired characteristics." Evaluate this claim.